

D 131477**(Pages : 3)****Name.....****Reg. No.....****THIRD SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2025****Mathematics****MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2****(2020–2023 Admissions)****Time : Two Hours and a Half****Maximum : 80 Marks****Section A**

*All questions can be attended.
Each question carries 2 marks.
Ceiling is 25.*

1. Differentiate the function $f(x) = \log\left(\frac{\log(x)}{x^2}\right)$.
2. Find the derivative of $y = \log(|\csc(x) - \cot(x)|)$.
3. Find $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x}$.
4. Show that $\cosh^{-1} x = \log(x + \sqrt{x^2 - 1})$.
5. Find $\lim_{n \rightarrow \infty} \frac{n+1}{\sqrt{n^2 + n + 1}}$.
6. Determine whether the sequence $a_n = \frac{\sin(1/n)}{n}$ converges or diverges. If it converges, find the limit.
7. Express $0.\overline{784784} \dots$ as a rational number.

Turn over

8. Show that the series $\sum_1^{\infty} (3)^{2n} 5^{1-n}$ is divergent.
9. By using the power series expansion of e^{-x} , show that $\frac{d}{dx}(e^{-x}) = -e^{-x}$.
10. Find the Taylor series expansion of $\sin x$ about $x = \pi/2$.
11. Find the parametric equation of the curve whose rectangular equation is $xy = 1$.
12. Show that the graph of the function $f(x) = (x-4)^{2/3}$ has a vertical tangent at $x = 4$.
13. Find the distance between the point $(1, -4, -3)$ and the plane $2x - 3y + 6z + 1 = 0$.
14. Find $r(t)$ given that $r'(t) = 3i + 2tj$ and $r(1) = 2i + 5j$.
15. Find the co-ordinates of point where the line $r(t) = (2+t)i + (1-2t)j + 3tk$ intersects the xz plane.

(Ceiling 25)

Section B

*All questions can be attended.
Each question carries 5 marks.
Overall Ceiling is 35*

16. Evaluate :

(i) $\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x}$.

(ii) $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin x}$.

17. Use Logarithmic differentiation to find the derivative of $y = \sqrt{x^x}$.
18. Find $\int \sin(x) e^x dx$.
19. Show that the sequence $a_n = \frac{n!}{n^n}$ converges and find its limit.

20. Find the interval of convergence and radius of convergence of the power series $\sum_0^{\infty} n!x^n$.
21. Find the Maclaurian series for $\sinh x$ and determine its interval of convergence.
22. Find the arc length of the cardioid $r = 1 + \cos \theta$.
23. Find the curvature and radius of curvature of the curve $r(t) = 2\sin(2t)i + \sin(t)j$ at $t = \pi/2$.

(Ceiling 35)

Section C

*Answer any two questions.
Each question carries 10 marks.*

24. (i) Evaluate $\int x^2 \log(x) dx$.
- (ii) Find $\int \frac{1}{\sqrt{1+e^{-2x}}} dx$.
- (iii) Find $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$.
25. (i) Let C be the circle $r(t) = \cos t + \sin t + 2k$. Find T(t) and N(t) at $t = \pi/4$.
- (ii) Identify and sketch the graph of the surface $z = \frac{y^2}{4} - \frac{x^2}{9}$.
26. (i) Find $\frac{d^2y}{dx^2}$ for $x = \sin(t), y = \sin(2t)$.
- (ii) Find the area of the region in the first quadrant that is within the cardioid $r = 1 - \cos \theta$.
27. (i) Find the equation of the paraboloid $z = x^2 + y^2$ in cylindrical and spherical co-ordinates.
- (ii) Find the spherical co-ordinates of the point that has rectangular co-ordinates $(4, -4, 4\sqrt{6})$.
- (2 × 10 = 20 marks)